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324355 (24)

BE (3rd Semester)

Examination, April-May, 2014

Branch : Electrical Engineering

LECTRICAL ENGG. MATERIAL (NEW)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

te : Solve any two parts from (b), (c) and (d) from each question. Part (a) is compulsory from each question.

1. (a) What are the materials used for fuses ? 2

(b) What do you mean by low resistivity materials ? Name any three materials and

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explain any one material with properties and uses. 7

(c) Explain the properties and uses in electrical industry of the following conducting materials. 7

(i) High conducting materials

(ii) High resistivity materials

(d) Explain the super conductors and also describe their properties and applications.

Q. 2. (a) What is cut-in voltage in semiconductor ? 2

(b) Describe the difference between p-type and n-type semiconductor materials. Sketch the

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atomic structure of silicon and insert the impurity of arsenic in silicon. 7

(c) What is Hall effect in semiconductor. Explain its origin and significance. Deduce an expression for Hall coefficient. 7

(d) What do you mean by "Fermi level" ? Derive the expression for fermi level and sketch the position in case of intrinsic p-type and m-type semiconductor. 7

3. (a) What do you understand by polarization ? 2

(b) How are solid dielectric material classified as per temperature limits ? Briefly discuss the

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features of materials in each class and na some materials in each class.

(c) What is meant by the terms dielectric lo and loss angle of a dielectric ? Deduc expressions for the same. Give the importance in dielectric materials.

(d) Write short notes on the followings

(i) Transformer oil

(ii) Mica based materials

(iii) Classification of solid insulating materials.

Q. 4. (a) What is Curie temperature ? 2

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(b) Explain the classification, properties and uses of ferromagnetic materials. 7

(c) Explain the phenomenon of magnetostriction and hysteresis in ferromagnetic materials. 7

(d) Write short notes on piezo-electric materials and their properties. 7

Q. 5. (a) What is mean by elastic limit ? 2

(b) What is photo-electric emission ? State and explain the laws of photo-electric emission ? 7

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(c) What is luminescence ? Explain the term photoluminescence, cathode luminescence and electro-luminescence. 7

(d) What do you mean by optical properties of semiconductor ? Define briefly. 7

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